

Semirings for database provenance

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What is a database?

What is database provenance?

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Provenance is typically represented as a general form of annotation information, usually by tagging data on the tuple-level or value-level of a relation.

What is a semiring?

Example

Edges

from	to	
<i>a</i>	<i>a</i>	p
<i>a</i>	<i>b</i>	q
<i>b</i>	<i>a</i>	r
<i>b</i>	<i>c</i>	s

Example

Query: Where can I get to in 3 hops?

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Query: Where can I get to in 3 hops?

Provenance: *Which* edges do I use?

I can get to (a, a) by using p three times.

I can also get to (a, a) by using p then q and then r .

Finally, I can get to (a, a) by using q then r and then p .

Semiring framework

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The provenance for (a, a) from our query from the previous slide would be the polynomial $p^3 + 2pqr$.

Positive relational algebra

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- Aggregation
- Difference

Aggregate queries

Books

title	genre	qty
book 1	adventure	4
book 2	fantasy	5
book 3	romance	6
book 4	adventure	3
book 5	fantasy	1
book 6	romance	2

GROUP BY genre

genre	total
adventure	7
fantasy	6
romance	8

SUM(qty)

Aggregation provenance

Difference in provenance

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- Monus operation on naturally ordered semirings
- Quotient constructions

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- Unified framework for recursion, aggregates, and difference
- Trade-offs between generality and usability
- Limits of semiring-based provenance